IB Computer Science Syllabus (2027) - Standard Level (Core)

This list covers the core content required for both Standard Level (SL) and Higher Level (HL) students.

Syllabus Point	Content Covered	Student Expected Outcome
A1.1.1	CPU components (ALU, CU, registers, buses)	**Describe** functions and interactions of CPU components.
A1.1.2	GPU (Graphics Processing Unit)	**Describe** the role of a GPU.
A1.1.4	Primary memory (RAM, ROM, Cache, Registers)	**Explain** the purposes of different primary memory types.
A1.1.5	Machine instruction cycle	**Describe** the fetch, decode, and execute cycle.
A1.1.7	Secondary storage (Internal/External)	**Describe** internal and external types of secondary storage.
A1.1.8	Compression	**Describe** the concept of compression.
A1.1.9	Cloud computing services	**Describe** different types of cloud services (SaaS, PaaS, IaaS).
A1.2.1	Data representation methods	**Describe** principal methods (binary, hex, decimal) of representing data.
A1.2.2	Binary data storage	**Explain** how binary is used to store data (text, media,

		etc.).
A1.2.3	Logic gates	**Describe** purpose and use of logic gates.
A1.2.4	Truth tables	**Construct and analyse** truth tables.
A1.2.5	Logic diagrams	**Construct** logic diagrams.
A1.3.1	Operating Systems (OS) role	**Describe** the role of operating systems.
A1.3.2	OS functions	**Describe** the functions of an operating system.
A1.3.3	Scheduling	**Compare** different approaches to scheduling.
A1.3.4	Polling vs. Interrupts	**Evaluate** the use of polling and interrupt handling.
A2.1.1	Network purpose & characteristics	**Describe** the purpose and characteristics of networks.
A2.1.2	Digital infrastructures	**Describe** purpose, benefits, and limitations of modern digital infrastructures.
A2.1.3	Network devices	**Describe** the function of network devices.
A2.1.4	Transport & App protocols	**Describe** network protocols used for transport and application.
A2.2.1	Network topologies	**Describe** functions and practical applications of topologies.
A2.2.3	Networking models	**Compare and contrast** networking models (Client-Server, P2P).

A2.2.4	Network segmentation	**Explain** concepts and applications of segmentation (VLANs).
A2.3.1	IP addressing	**Describe** different types of IP addressing (IPv4/v6, Static/Dynamic).
A2.3.2	Transmission media	**Compare** types of media for data transmission (Wired/Wireless).
A2.3.3	Packet switching	**Explain** how packet switching is used to send data.
A2.4.1	Firewalls	**Discuss** the effectiveness of firewalls.
A2.4.4	Encryption & Digital Certificates	**Describe** the process of encryption and digital certificates.
A3.1.1	Relational databases	**Explain** features, benefits, and limitations of relational databases.
A3.2.1	Database schemas	**Describe** database schemas.
A3.2.2	ERDs (Entity Relationship Diagrams)	**Construct** ERDs.
A3.2.3	Data types	**Outline** different data types in relational databases.
A3.2.4	Tables	**Construct** tables for relational databases.
A3.2.5	Normal forms	**Explain** the difference between normal forms (1NF, 2NF, 3NF).

A3.2.6	Normalization (3NF)	**Construct** a database normalized to 3NF for real scenarios.
A3.2.7	Denormalization	**Evaluate** the need for denormalizing databases.
A3.3.1	SQL languages (DDL/DML)	**Outline** differences between data language types in SQL.
A3.3.2	SQL Queries	**Construct** queries between two tables in SQL.
A3.3.3	SQL Updates	**Explain** how SQL is used to update data.
A4.1.1	ML types & applications	**Describe** types of machine learning and real-world applications.
A4.1.2	ML hardware requirements	**Describe** hardware requirements for ML scenarios.
A4.4.1	Ethical implications of ML	**Discuss** ethical implications of ML in real-world scenarios.
A4.4.2	Ethics of tech integration	**Discuss** ethical aspects of increasing tech integration in daily life.
B1.1.1	Problem specification	**Construct** a problem specification.
B1.1.2	Computational thinking concepts	**Describe** fundamental concepts (Abstraction, Decomposition, etc.).
B1.1.3	Problem solving	**Explain** how computational thinking is used to solve CS problems.

B1.1.4	Flowcharts	**Trace** flowcharts for programming algorithms.
B2.1.1	Variables (Global/Local)	**Construct and trace** programs using global/local variables.
B2.1.2	Substrings	**Construct** programs that extract and manipulate substrings.
B2.1.3	Exception handling	**Describe** how programs use exception handling techniques.
B2.1.4	Debugging	**Construct and use** common debugging techniques.
B2.2.1	Static vs Dynamic data structures	**Compare** static and dynamic data structures.
B2.2.2	Arrays and Lists	**Construct** programs that apply arrays and Lists.
B2.2.3	Stacks (LIFO)	**Explain** the concept of a stack.
B2.2.4	Queues (FIFO)	**Explain** the concept of a queue.
B2.3.1	Sequence constructs	**Construct** programs implementing correct instruction sequences.
B2.3.2	Selection constructs	**Construct** programs utilizing selection structures (if/else).
B2.3.3	Looping constructs	**Construct** programs utilizing looping structures.
B2.3.4	Functions & Modularization	**Construct** functions and modularization.

B2.4.1	Algorithm efficiency (Big O)	**Describe** efficiency of algorithms by calculating Big O.
B2.4.2	Linear & Binary Search	**Construct and trace** linear and binary search algorithms.
B2.4.3	Bubble & Selection Sort	**Construct and trace** bubble and selection sort algorithms.
B2.5.1	File processing	**Construct** code to perform file-processing operations.
B3.1.1	OOP Fundamentals	**Evaluate** the fundamentals of OOP.
B3.1.2	Class design	**Construct** a design of classes, methods, and behaviour.
B3.1.3	Static vs Non-static	**Distinguish** between static and non-static variables/methods.
B3.1.4	Class definition & Instantiation	**Construct** code to define classes and instantiate objects.
B3.1.5	Encapsulation	**Explain and apply** encapsulation and information hiding.